

IB MATH AA SL

Lesson 18

The Unit Circle & Exact Values

Trigonometry · 60-minute lesson

Syllabus SL 3.5 · Property of Lynda Badmus Education

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Do Now — recall

ON YOUR WHITEBOARD

1. What is $\sin 90^\circ$?
2. What is $\cos 0^\circ$?
3. On the unit circle, what are the coordinates at angle 0° ?

Answers (click to reveal)

1. 1
2. 1
3. (1, 0)

What we're learning today

LEARNING OBJECTIVES

- 1 Use**
the unit circle to define sin, cos and tan.
- 2 Read**
exact values for key angles.
- 3 Determine**
the sign of a ratio in each quadrant.
- 4 Find**
exact values without a calculator.

SUCCESS CRITERIA — I CAN...

- ✓
link $(\cos \theta, \sin \theta)$ to a point on the unit circle
- ✓
quote exact values for $0, 30, 45, 60, 90^\circ$
- ✓
use CAST to get the sign in each quadrant
- ✓
find exact ratios for related angles

The unit circle

DEFINITIONS

$$x = \cos \theta, \quad y = \sin \theta, \quad \tan \theta = \frac{\sin \theta}{\cos \theta}$$

Point

$(\cos \theta, \sin \theta)$ on a circle of radius 1

tan

the gradient of the radius line

CAST (signs)

Quadrant 1

All positive

Quadrant 2

Sin positive

Quadrant 3

Tan positive

Quadrant 4

Cos positive

Exact value

QUESTION

Find the exact value of $\sin 30^\circ$ and $\cos 30^\circ$.

SOLUTION

From the 30–60–90 triangle:

$$\sin 30^\circ = \frac{1}{2}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

These are exact — no decimals.

Your turn — exact values

ANSWER IN YOUR BOOK

1. Write $\cos 60^\circ$.
2. Write $\sin 45^\circ$.
3. Write $\tan 45^\circ$.

Answers (click to reveal)

1. $\frac{1}{2}$
2. $\frac{\sqrt{2}}{2}$
3. 1

Signs & related angles

USING SYMMETRY

$$\sin (180^\circ - \theta) = \sin \theta$$

Use CAST to decide + or -

Relate the angle to its acute 'reference' angle

EXAMPLES

$$\begin{aligned}\sin 150^\circ \\ &= \sin 30^\circ = 1/2\end{aligned}$$

$$\begin{aligned}\cos 120^\circ \\ &= -\cos 60^\circ = -1/2\end{aligned}$$

$$\begin{aligned}\tan 225^\circ \\ &= \tan 45^\circ = 1\end{aligned}$$

Sign from the quadrant

QUESTION

Find the exact value of $\cos 120^\circ$.

SOLUTION

120° is in Quadrant 2 $\rightarrow$ cosine is **NEGATIVE**:

Reference angle is $180^\circ - 120^\circ = 60^\circ$:

$$\cos 120^\circ = -\cos 60^\circ$$

So:

$$= -\frac{1}{2}$$

Exam-style question

- (a) Write down the exact value of $\sin 60^\circ$. [1]
(b) Find the exact value of $\cos 150^\circ$. [3]

MARK SCHEME — reveal after students attempt (tutor only)

- (a) $\sqrt{3}/2$ (A1)
(b) Q2 $\rightarrow$ cos negative; ref 30° (M1) $\rightarrow \cos 150^\circ = -\cos 30^\circ$ (A1) $= -\sqrt{3}/2$ (A1)

Common mistakes



Mode

Be clear whether you are working in degrees or radians.



Quadrant sign

Use CAST — sin/cos/tan are not always positive.



Exact means surd

$\sqrt{3}/2$ is exact; 0.866 is not.



$\tan = \sin/\cos$

$\tan$ is undefined where $\cos = 0$ (90° , 270°).

Mixed practice

QUESTIONS

1. Exact value of $\tan 60^\circ$.
2. Exact value of $\sin 120^\circ$.
3. Exact value of $\cos 180^\circ$.
4. In which quadrant is only $\sin$ positive?

Answers (click to reveal)

1. $\sqrt{3}$
2. $\sqrt{3}/2$
3. -1
4. Quadrant 2

Plenary & exit ticket

KEY FORMULAS

$$x = \cos \theta, \quad y = \sin \theta$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

EXIT TICKET — before you leave

1. $\sin 90^\circ$?
2. Exact value of $\cos 45^\circ$?
3. Sign of $\sin$ in Quadrant 3?

Answers (click to reveal)

1. 1
2. $\sqrt{2}/2$
3. negative

Homework

SET ON DR FROST MATHS (auto-marked)

Task:

Unit circle & exact values.

- Aim for 80%+ before the next lesson
- Re-attempt any flagged questions
- Bring one tricky question to discuss

NEXT LESSON

Lesson 19

Trigonometric identities — relationships you can rely on.

$$\sin^2 \theta + \cos^2 \theta = 1$$